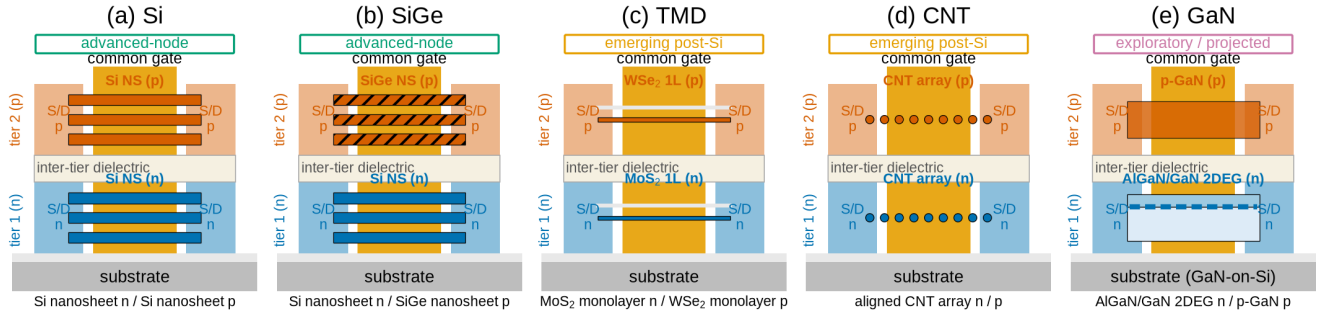


# Figures

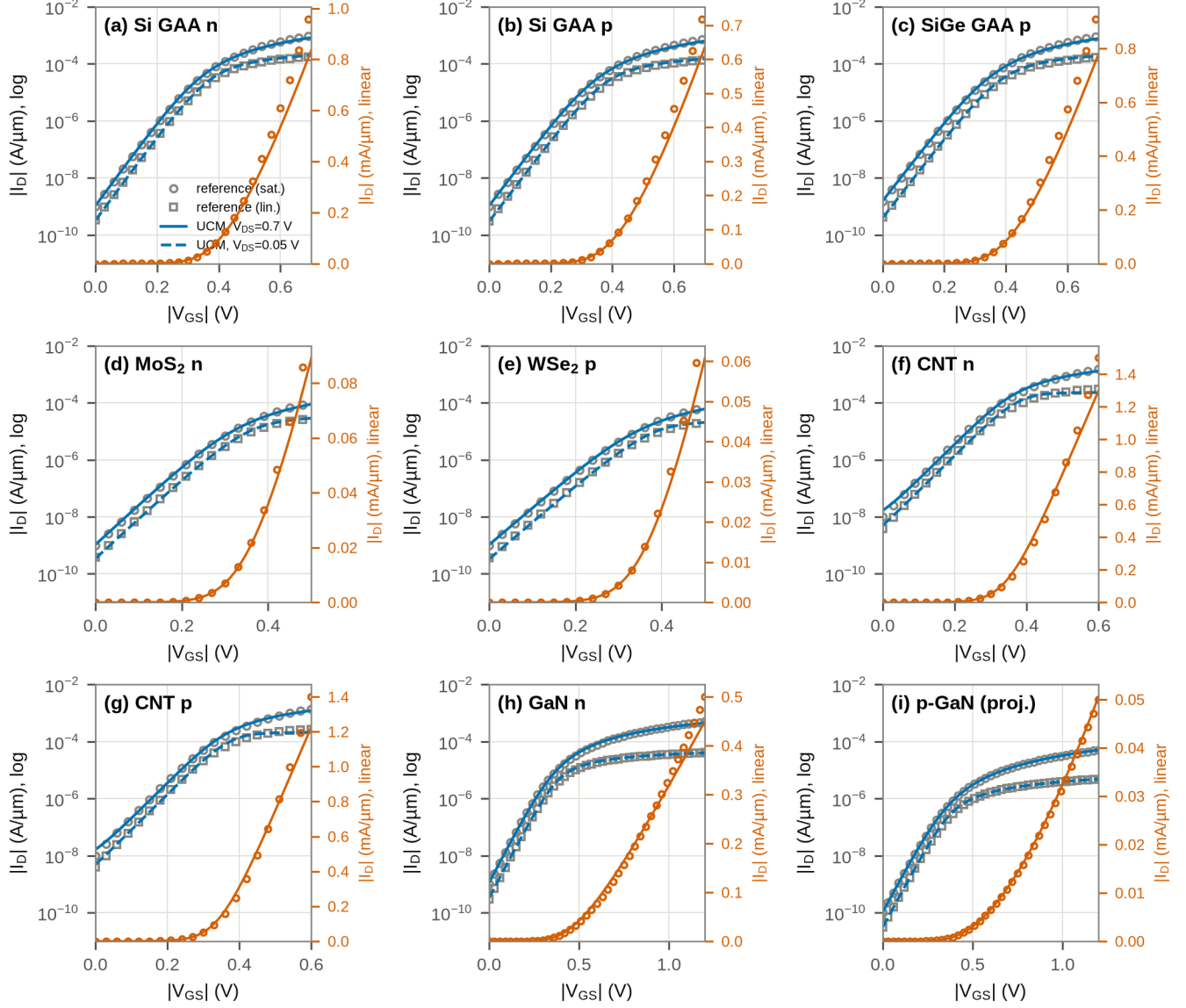
A Unified Physics-Based Compact Model and Calibrated Circuit-Validation Flow for Multi-Material CFET Logic:  
Si, SiGe, MoS<sub>2</sub>/WSe<sub>2</sub>, CNT and GaN

M. Nisha Angeline

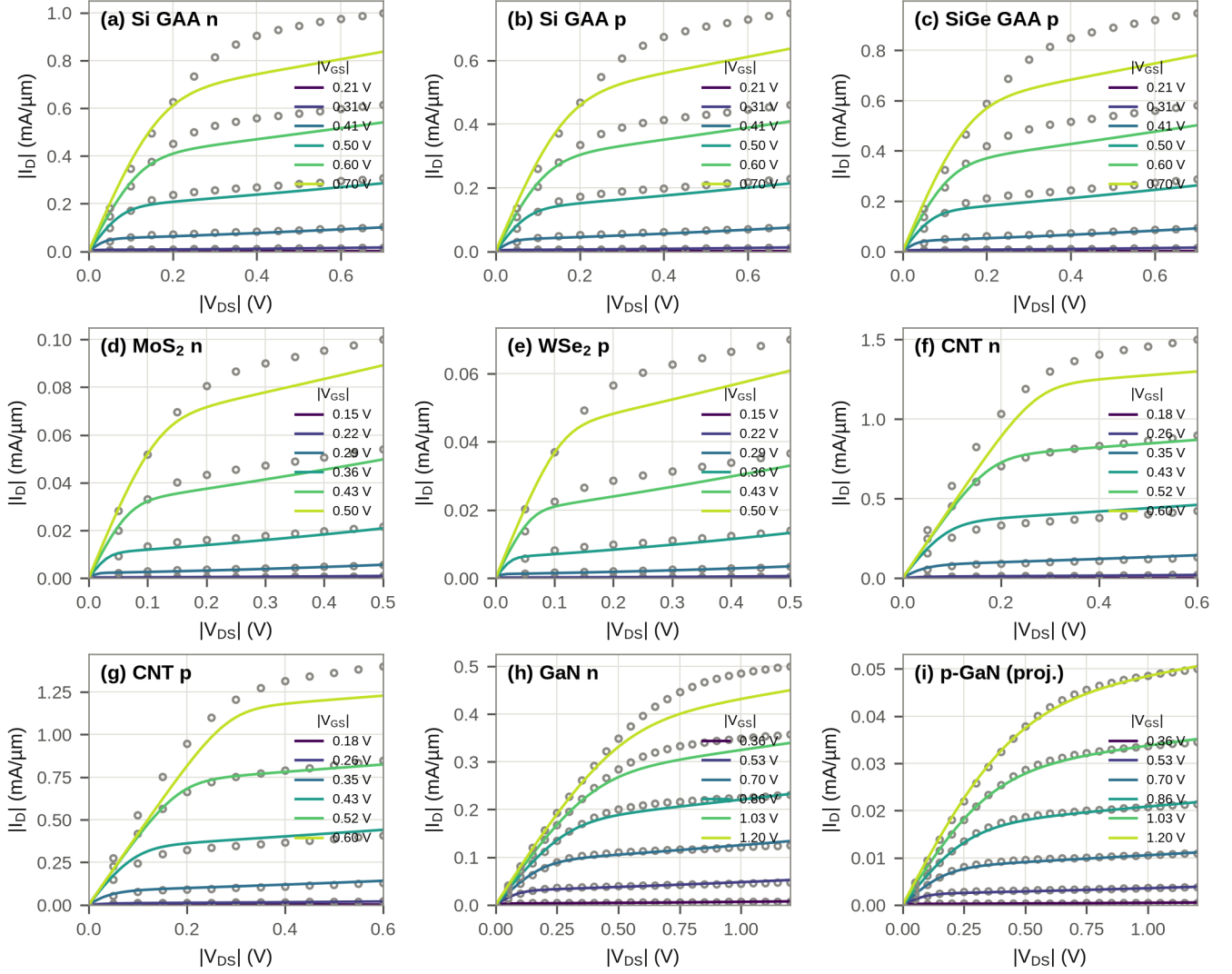
All figures are also provided as individual PNG (300 dpi) and PDF files in **figures/**, **schematics/** and **layout/png/**.



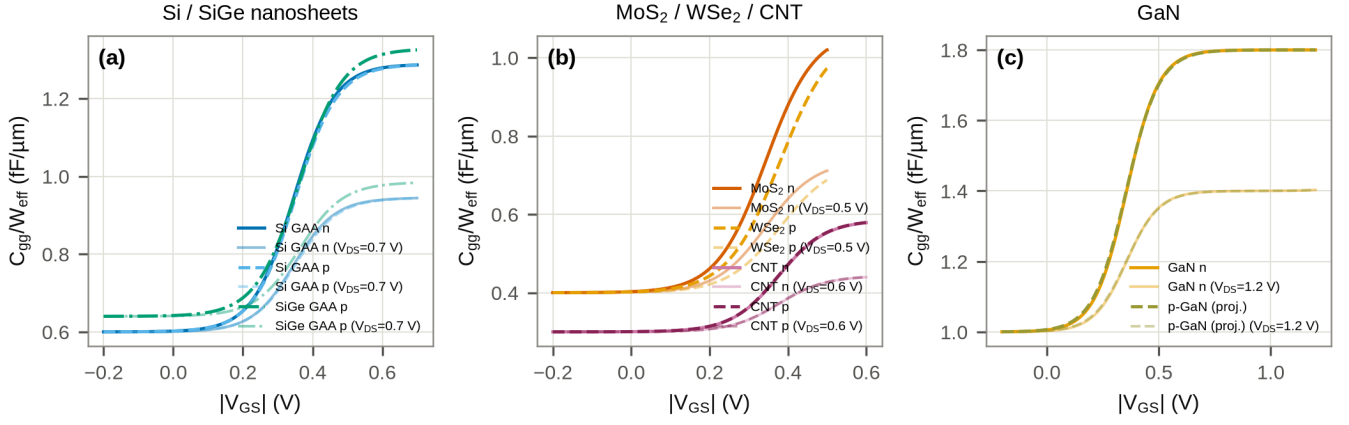
**Fig. 1.** Cross-section cartoons of the five CFET platforms: bottom n-tier, inter-tier dielectric, top p-tier, common gate wrapping both tiers, and n/p source/drain contacts. Maturity tags follow Table I.



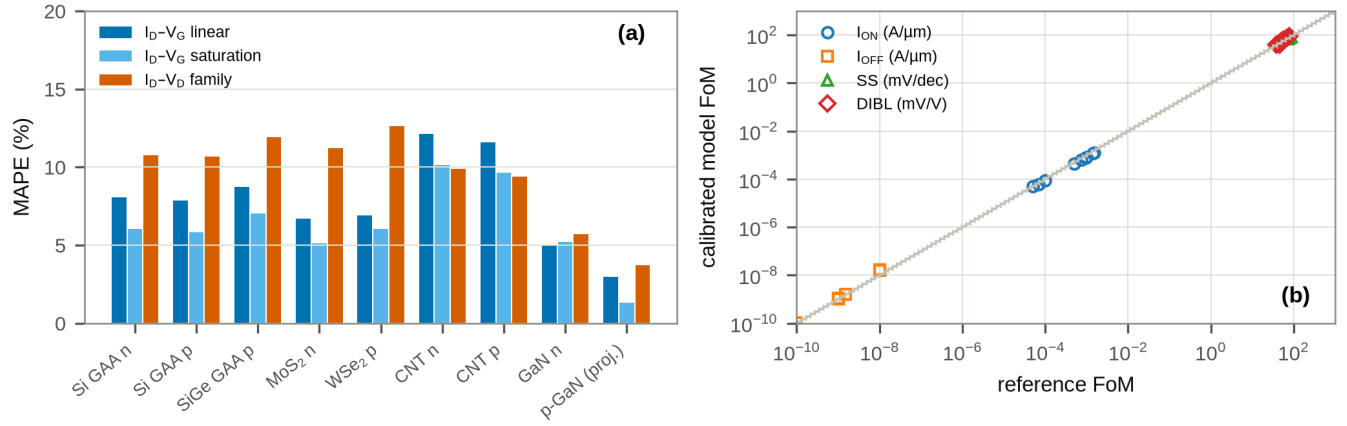
**Fig. 2.**  $I_D$ - $V_G$  calibration of the nine device models (lines) against the literature-anchored reference curves (symbols) at  $|V_{DS}| = 0.05$  V and  $|V_{DS}| = V_{DD}$ , on logarithmic (left axes) and linear (right axes) scales.



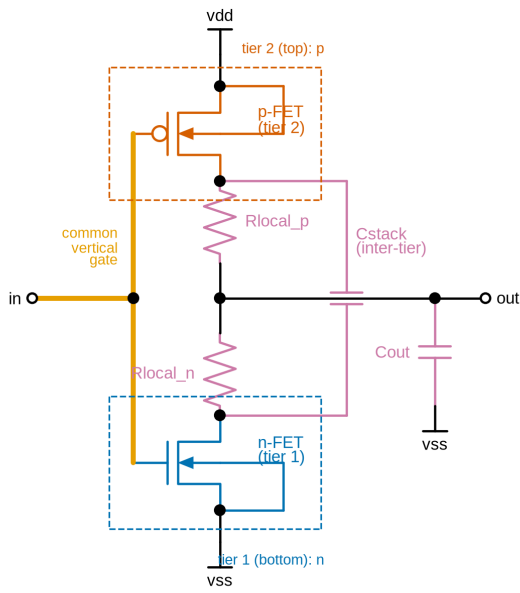
**Fig. 3.**  $I_D$ - $V_D$  families (six gate biases from  $0.3V_{DD}$  to  $V_{DD}$ ) of the calibrated models (lines) and reference (symbols).



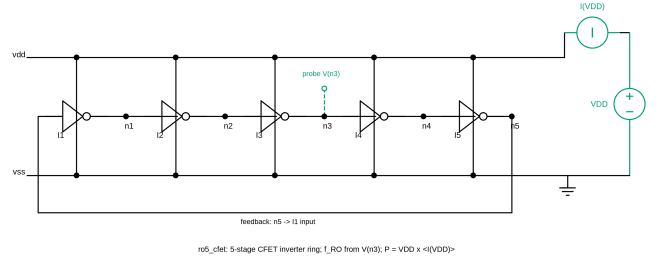
**Fig. 4.** Gate capacitance per effective width,  $C_{gg} = dQ_g/dV_{GS}$ , of the nine calibrated models from the charge model at  $V_{DS} = 0$  (full lines) and  $V_{DS} = V_{DD}$  (faded): (a) Si/SiGe nanosheets, (b) 2D and CNT channels, (c) GaN.



**Fig. 5.** (a) MAPE per device and data set; (b) parity of model vs. reference figures of merit.

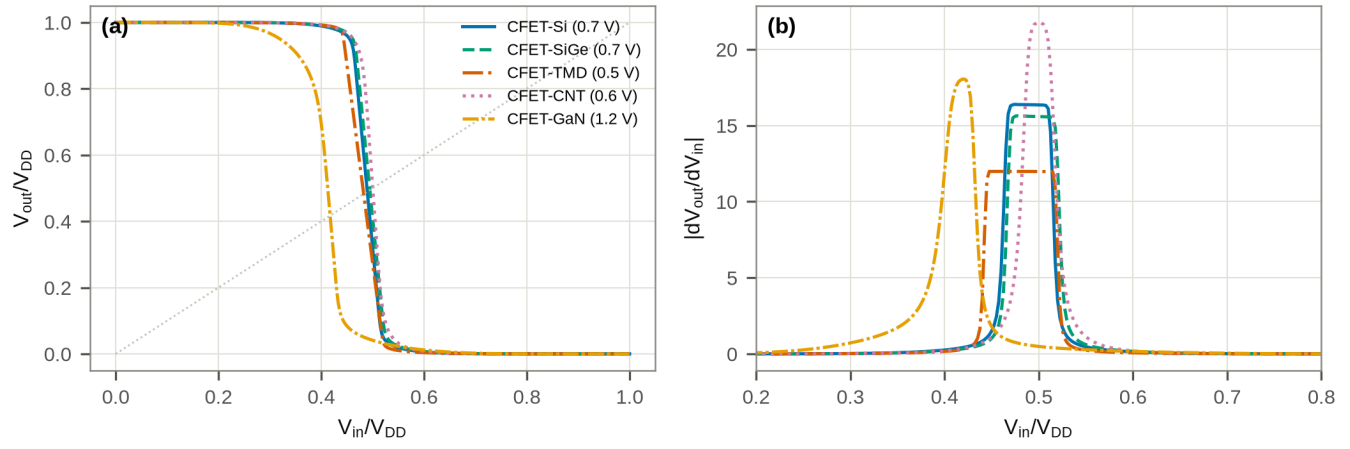


inv\_cfet: p-FET stacked over n-FET on one vertical gate; parasitics Rlocal\_n/p, Cstack, Cout

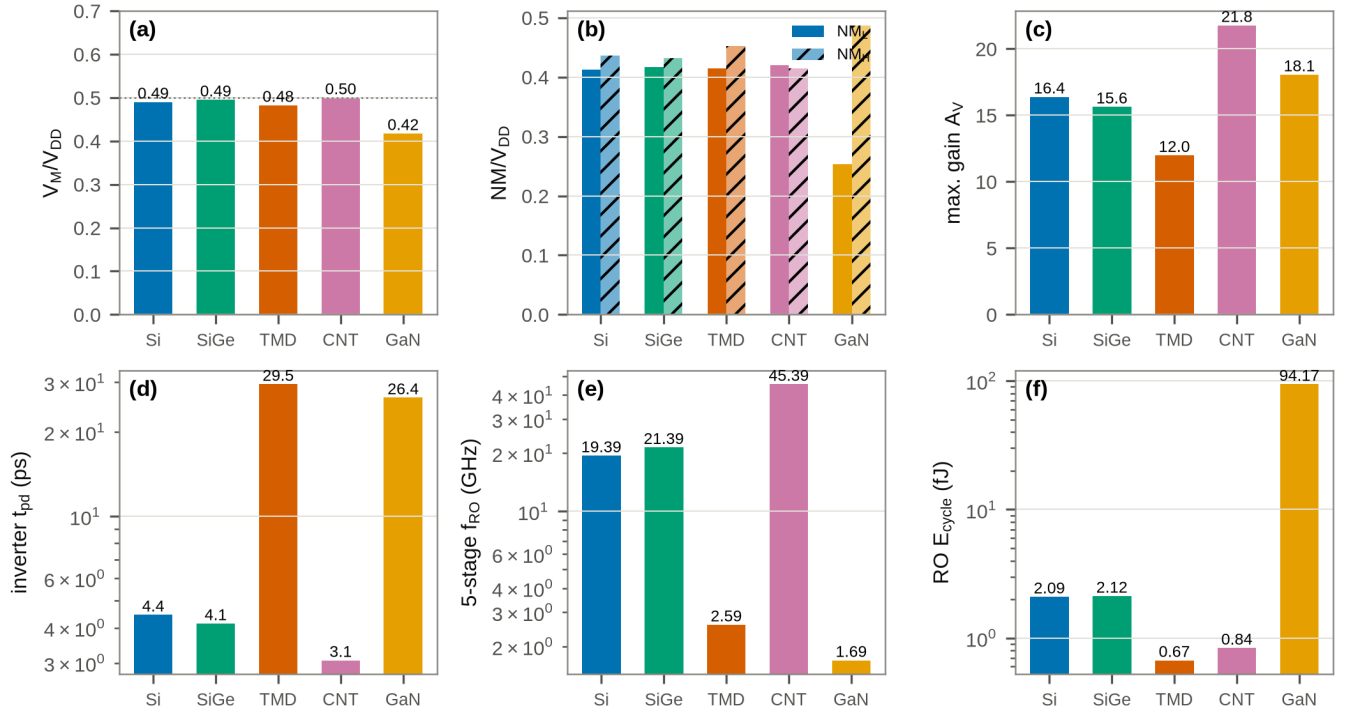


ro5\_cfet: 5-stage CFET inverter ring; I\_RD from V(n3); P = VDD x <I(VDD)>

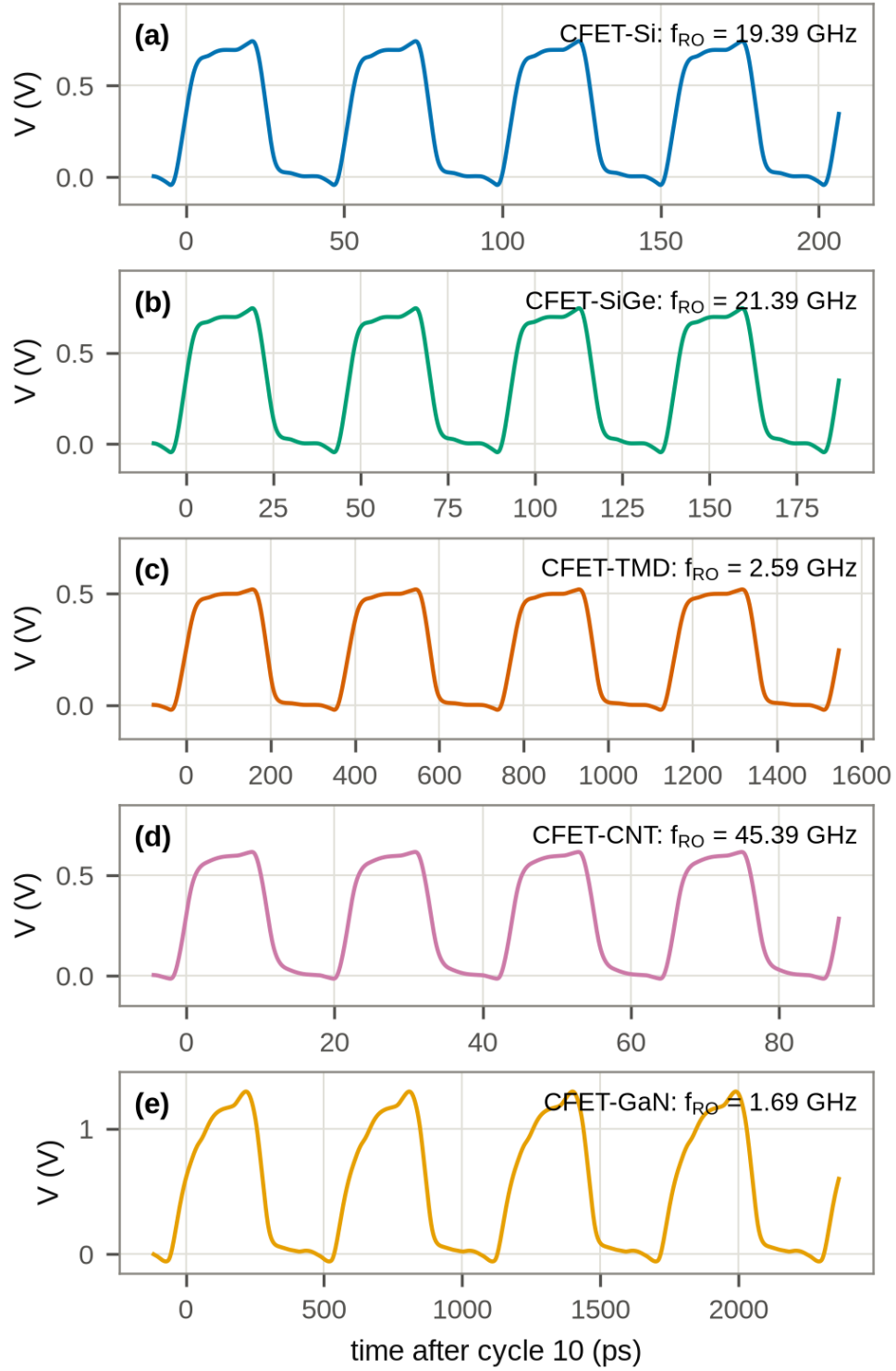
**Fig. 6.** (a) CFET inverter cell with the explicit inter-tier coupling capacitance  $C_{stack}$ , output parasitic  $C_{out}$  and local-interconnect resistances  $R_{local,n/p}$ ; (b) 5-stage ring oscillator testbench with the  $V(n_3)$  probe and the  $I(V_{DD})$  meter.



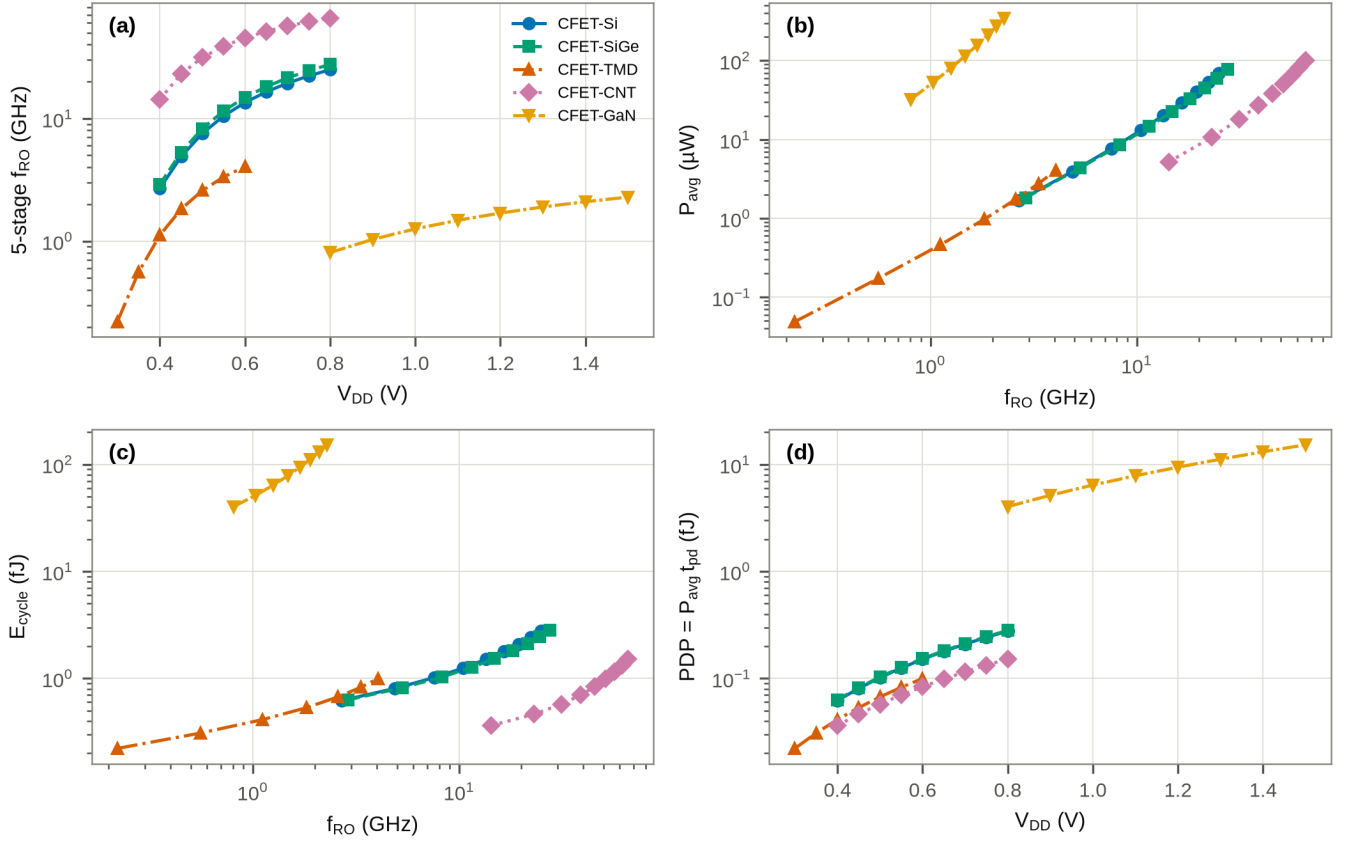
**Fig. 7.** (a) Normalised voltage-transfer characteristics of the five CFET inverters at their nominal  $V_{DD}$ ; (b) small-signal gain.



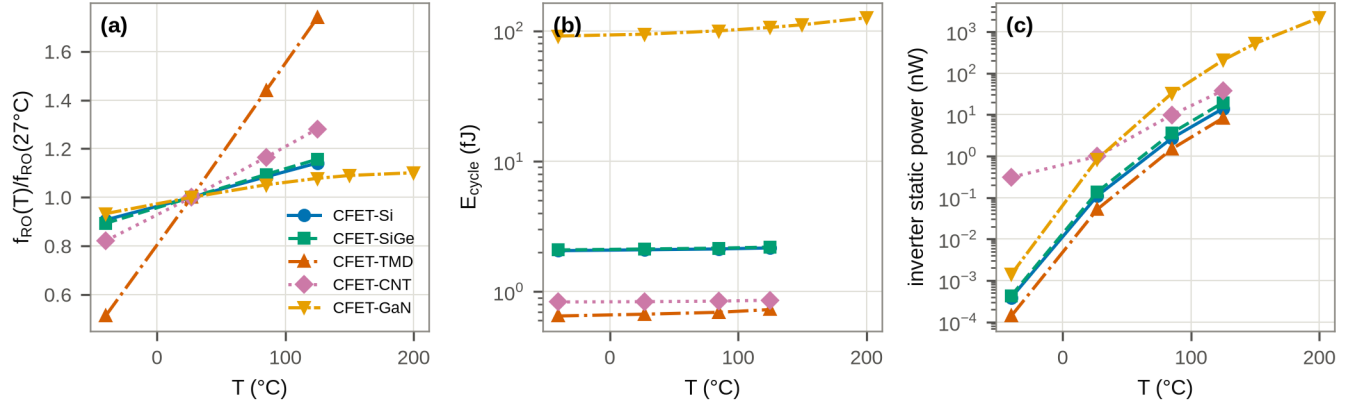
**Fig. 8.** Nominal figures of merit: (a)  $V_M/V_{DD}$ , (b) noise margins, (c) maximum gain, (d) inverter delay, (e) 5-stage RO frequency, (f) energy per RO cycle.



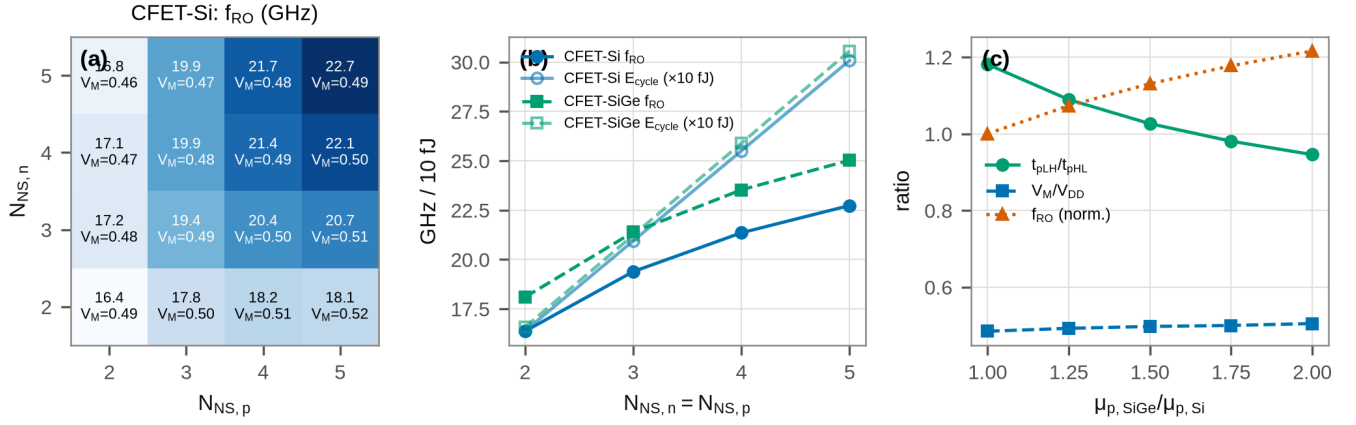
**Fig. 9.** 5-stage RO waveforms  $V(n_3)$  for the five platforms, four cycles after cycle 10 (the measurement window).



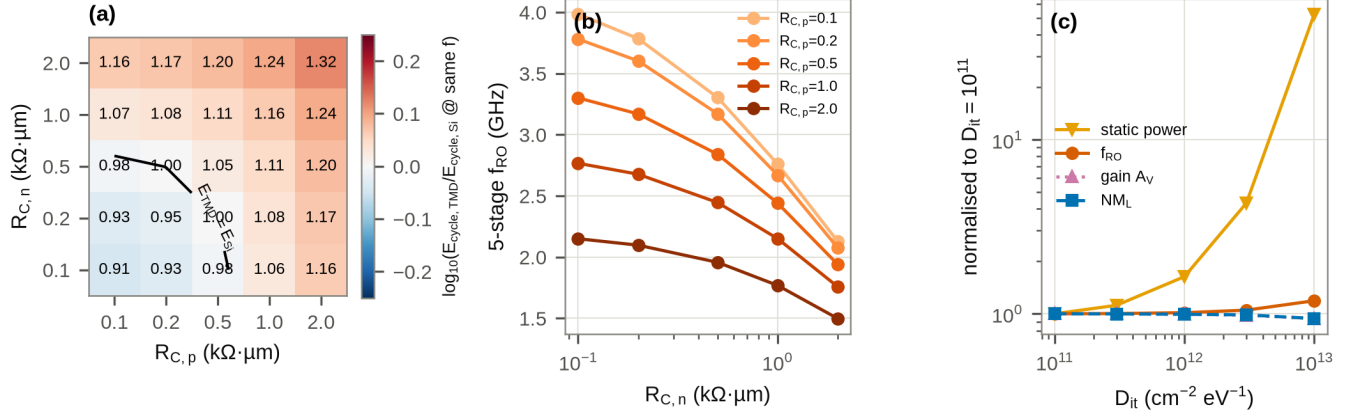
**Fig. 10.** Unified  $V_{DD}$  sweeps of the 5-stage ROs: (a)  $f_{RO}$  vs.  $V_{DD}$ , (b) power vs. frequency, (c) energy per cycle vs. frequency, (d) PDP vs.  $V_{DD}$ .



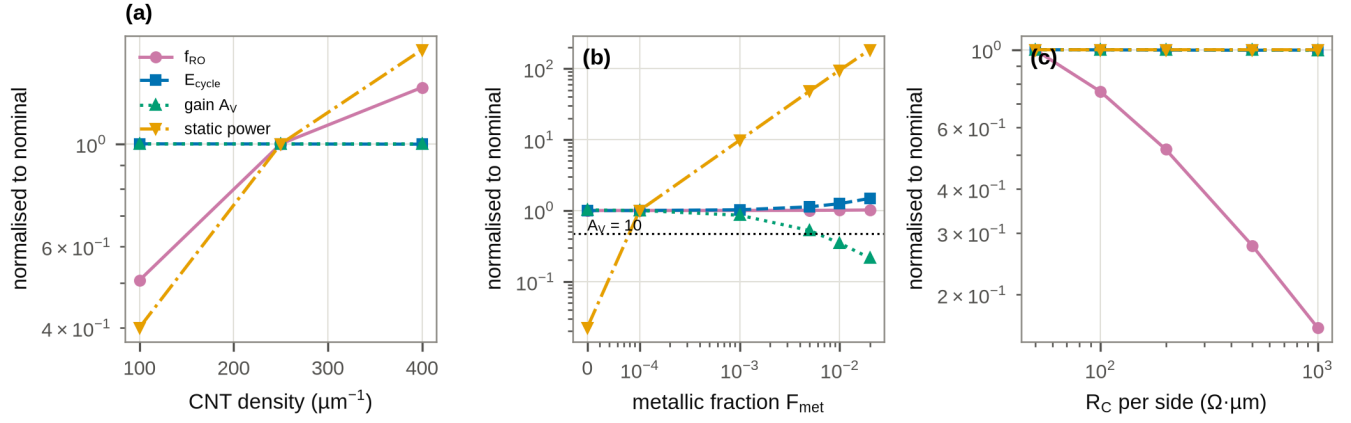
**Fig. 11.** Temperature sweeps at nominal  $V_{DD}$  ( $-40$  to  $125^{\circ}\text{C}$ , GaN to  $200^{\circ}\text{C}$ ): (a) normalised  $f_{RO}$ , (b) energy per cycle, (c) inverter static power.



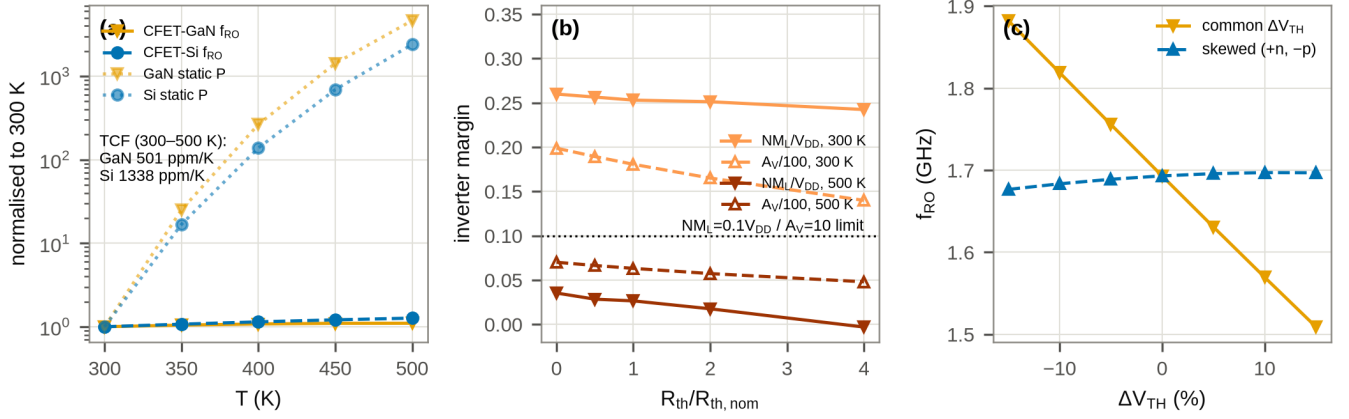
**Fig. 12.** Si/SiGe study: 5-stage  $f_{RO}$  and  $V_M/V_{DD}$  vs. the number of nanosheets in the n- and p-tier for (a) CFET-Si and (b) CFET-SiGe; (c) effect of the SiGe/Si hole-mobility ratio on the rise/fall delay mismatch,  $V_M$  and  $f_{RO}$ .



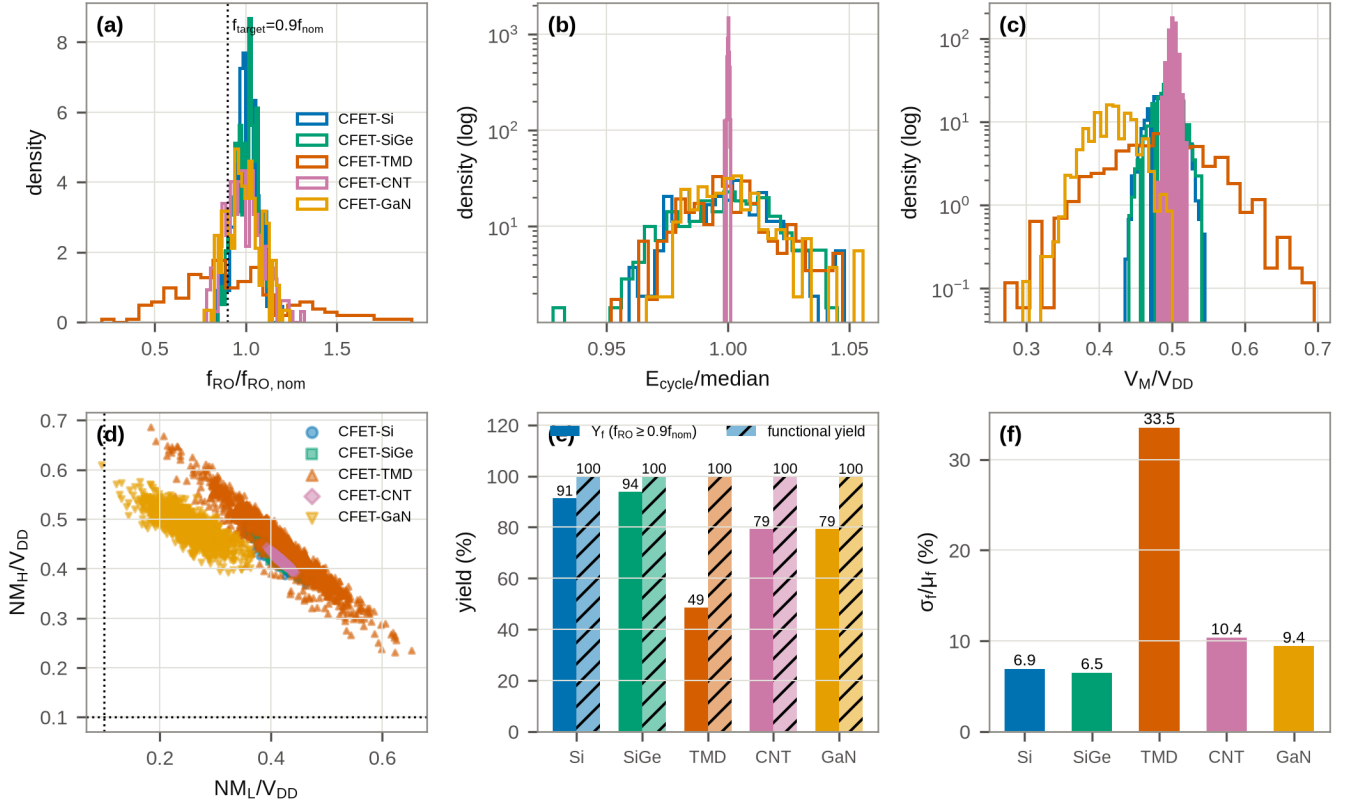
**Fig. 13.** TMD study: (a) iso-frequency energy-per-cycle ratio  $E_{TMD}/E_{Si}$  vs. the n- and p-contact resistance (the Si energy is interpolated at the TMD frequency from the Si  $V_{DD}$  sweep extended to 0.3 V); (b)  $f_{RO}$  vs.  $R_{C,n}$ ; (c) effect of the interface trap density.



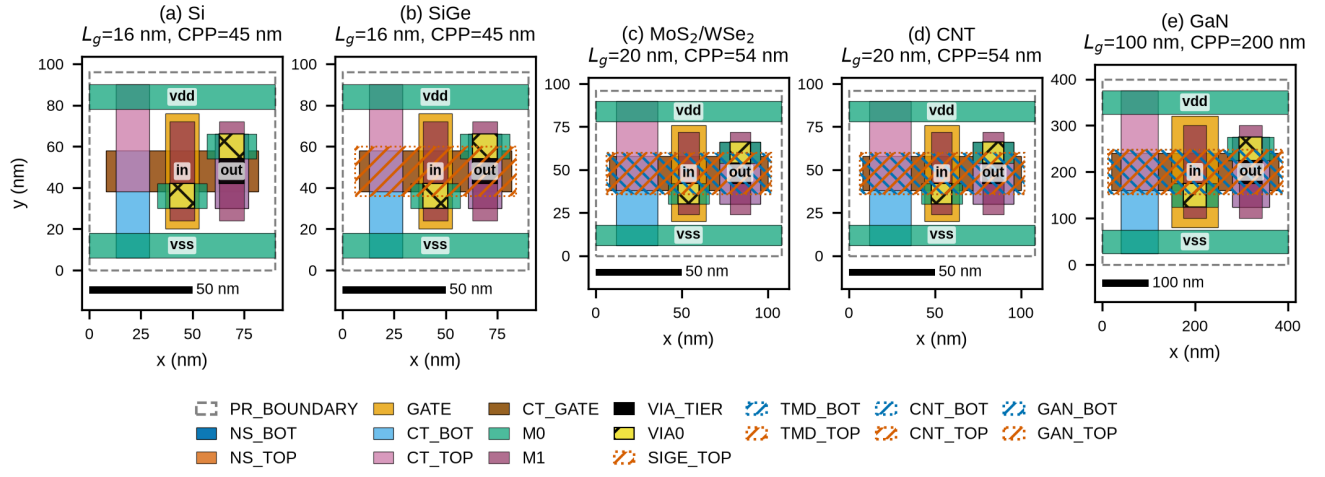
**Fig. 14.** CNT study: normalised  $f_{RO}$  and PDP (left axes) and inverter static power (right axes) vs. (a) tube density, (b) metallic-tube fraction, (c) contact resistance.



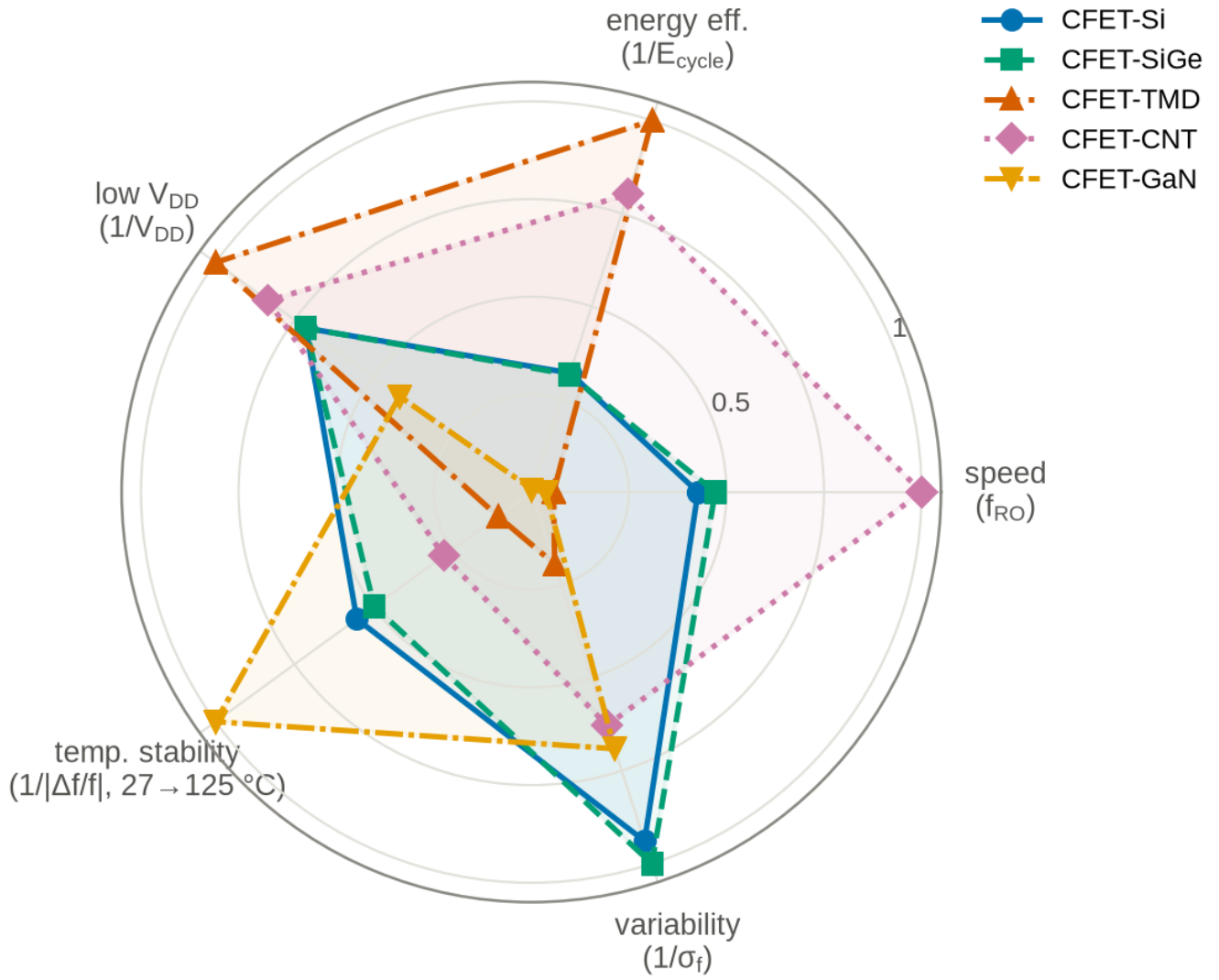
**Fig. 15.** GaN study: (a)  $f_{RO}$  and static power of CFET-GaN and CFET-Si from 300 to 500 K; (b) self-heating delay penalty vs. thermal resistance at three temperatures; (c)  $f_{RO}$  vs. common and skewed threshold-voltage shifts.



**Fig. 16.** Monte Carlo (200 RO samples and 1000 inverter samples per platform): (a)  $f_{RO}$  normalised to nominal, (b) energy per cycle normalised to its median, (c)  $V_M/V_{DD}$ , (d) noise-margin scatter with the  $0.1V_{DD}$  functional limits, (e) frequency yield  $Y_f$  and functional yield, (f) relative frequency spread.



**Fig. 17.** Conceptual, DRC-clean GDSII layouts of the five CFET inverter cells (plan view; tiers drawn as separate layers): Si and SiGe at  $L_g = 16$  nm, CPP = 45 nm, 90×96 nm cell; TMD and CNT projected at  $L_g = 20$  nm, CPP = 54 nm; GaN projected at  $L_g = 100$  nm, CPP = 200 nm.



**Fig. 18.** Material-selection chart: speed, energy efficiency, low- $V_{DD}$  capability, temperature stability ( $1/|\Delta f/f|$  between  $27$  and  $125^\circ\text{C}$ ) and variability ( $1/\sigma_f$ ), each normalised to the best platform.